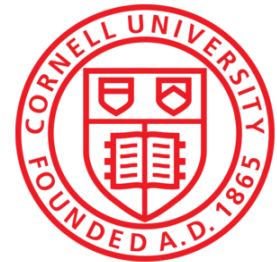


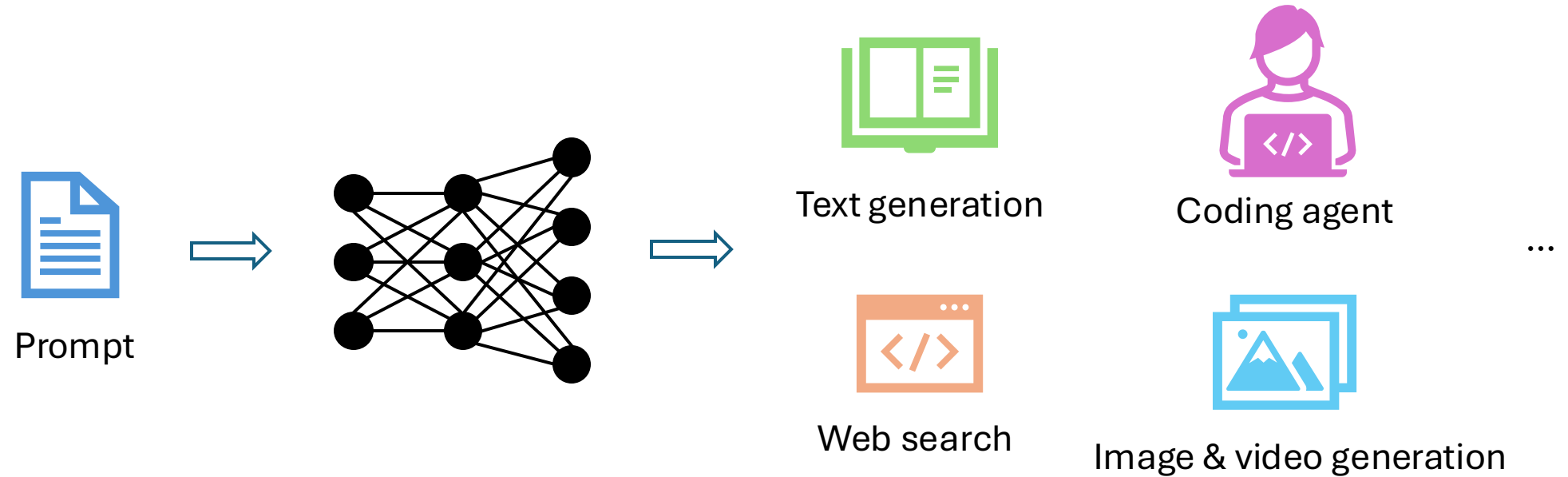
Photonic Rail Network in ML Datacenters

Eric Ding, Chuhan Ouyang, Rachee Singh
Cornell University



HotNets'25, Nov 17-18 2025, College Park, MD, USA

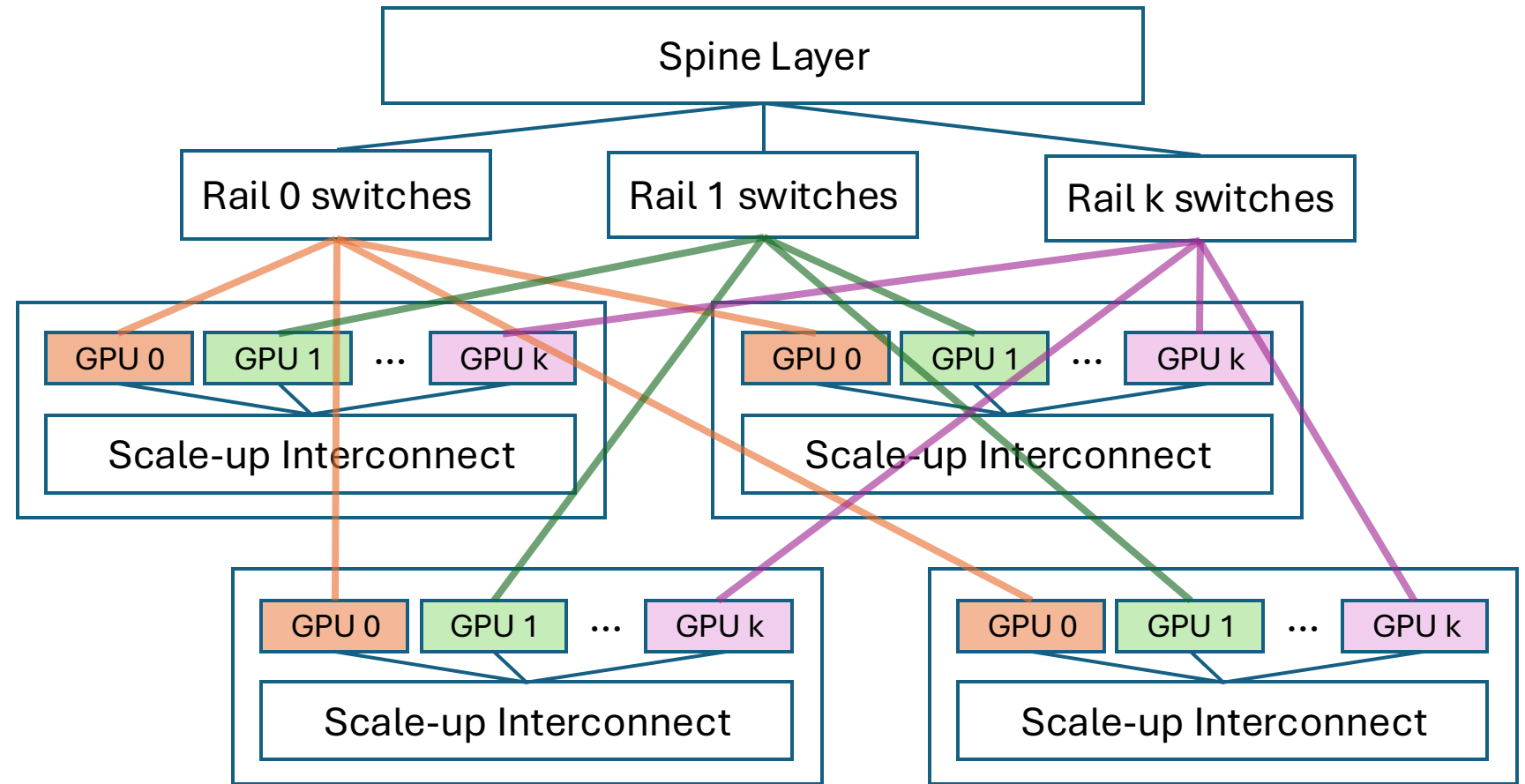
Gen AI is everywhere



To train / deploy these models, we are throwing more and more compute

Network bandwidth becomes bottleneck

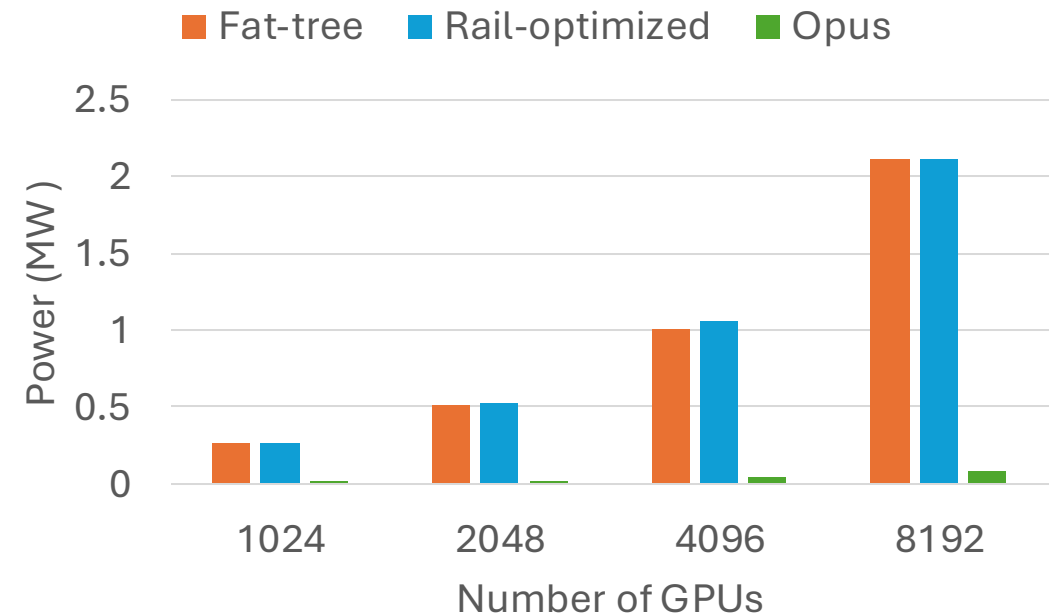
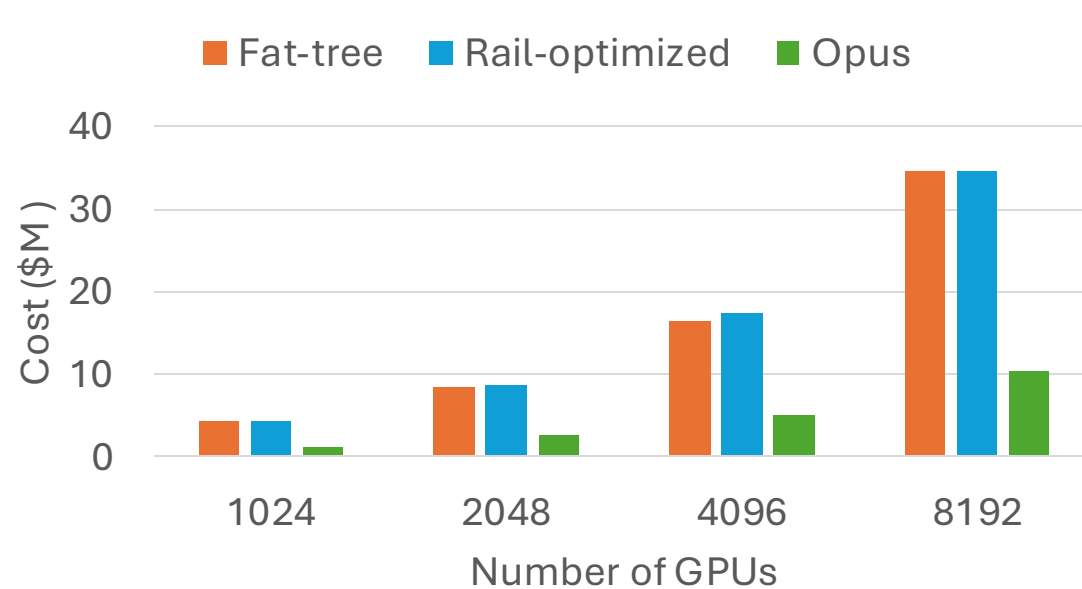
- State-of-the-art **scale-out** network uses parallel “rails”
- High-bandwidth electrical switches are deployed
- Cost and power



Rail-optimized network

CapEx and Energy of GPU scale-out network

DGX H200, 400Gbps transceivers and Tomahawk-4 64×400GbE switches [1-2]



Reduce cost by up to **70%** and power consumption by up to **96%**!

How can we achieve this?

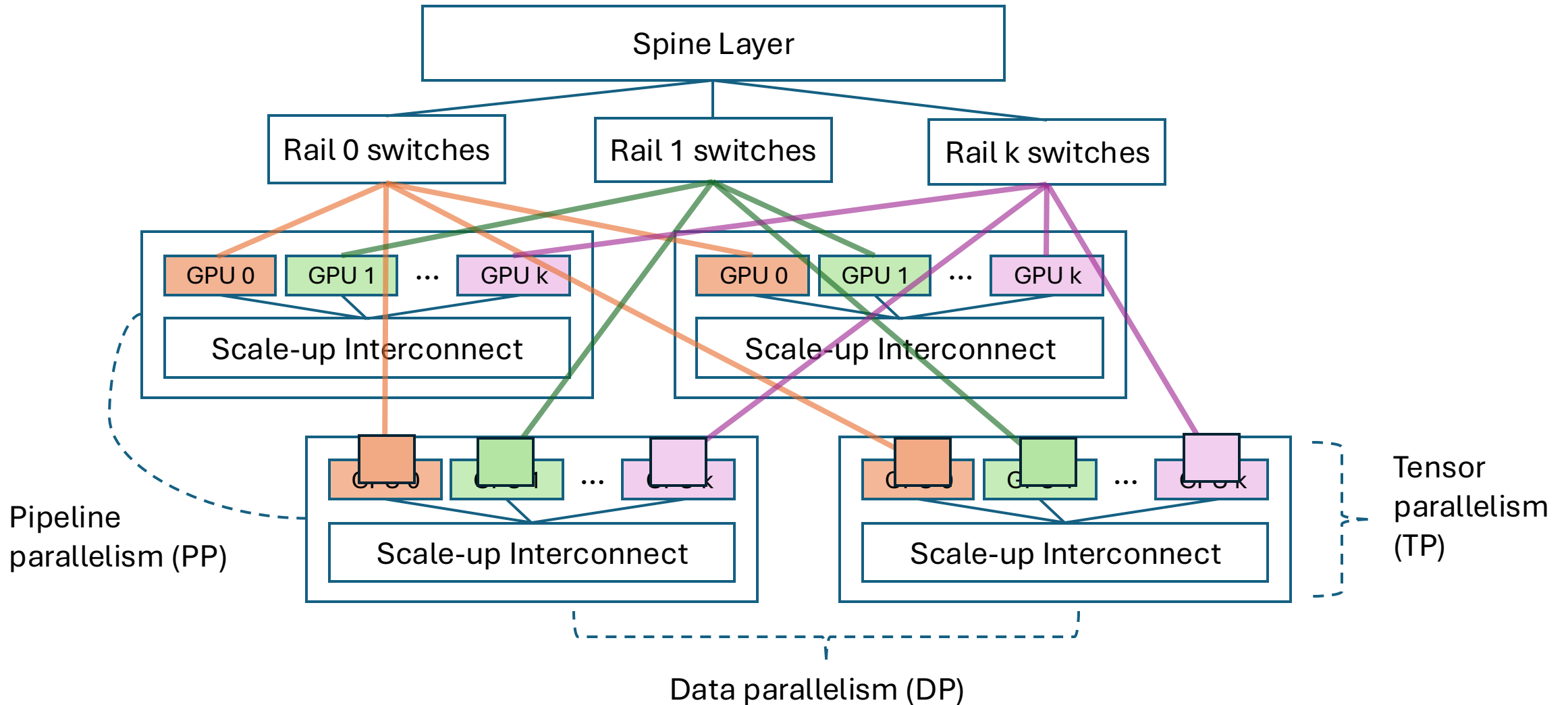
Dynamically align the network connectivity/bandwidth with the **active** communication groups within a workload (vs. static [4-5])

Dynamic bandwidth allocation →

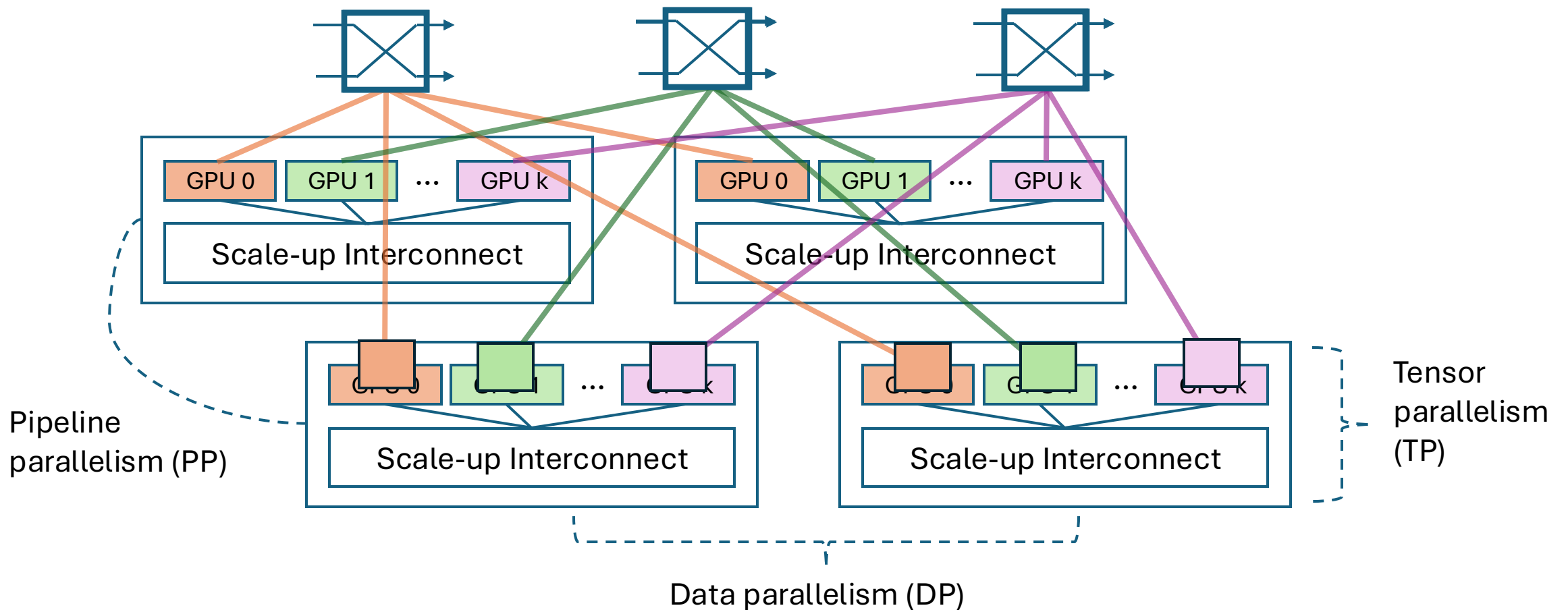
Reduce network components →

Less cost and energy consumption

Our proposal

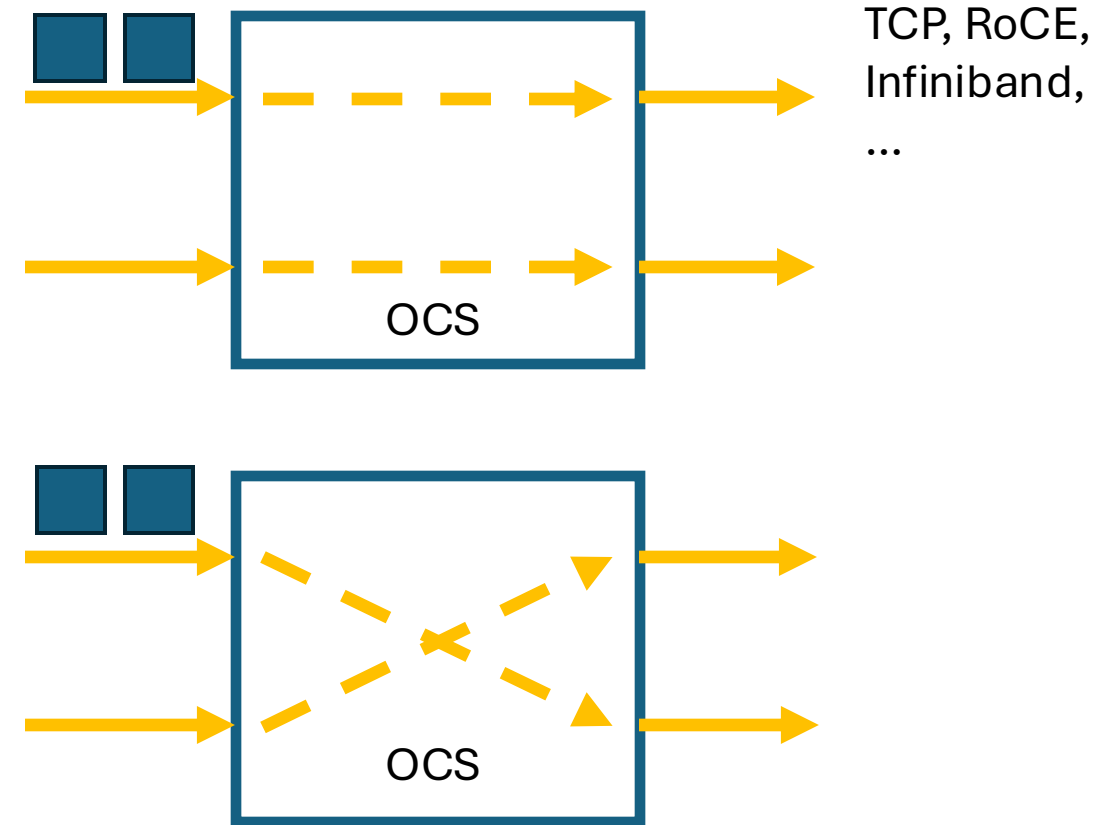


Our proposal: circuit switching



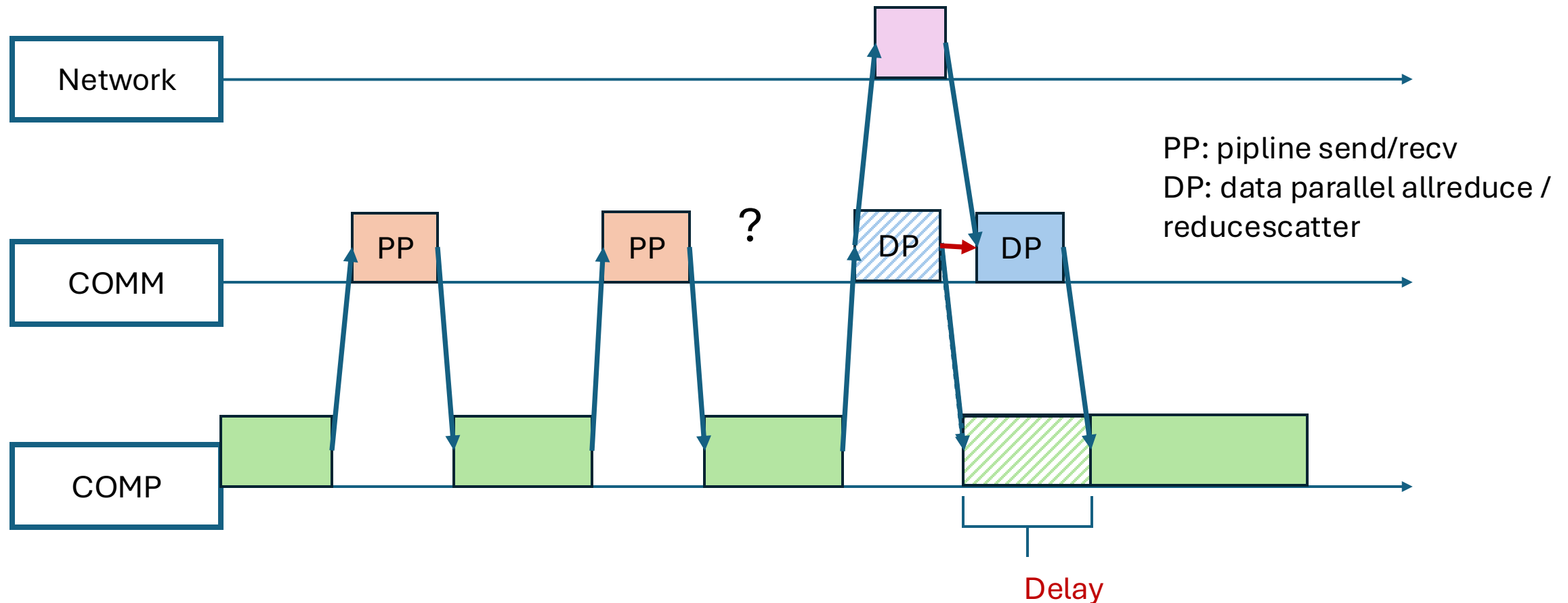
Optical circuit switching (OCS)

- Low latency 🍷
 - Packets remain in optical domain
 - No queueing and header processing
- Bandwidth-agnostic 🍷
 - Only bottleneck is transceiver bandwidth at the endpoints
- Reconfigure on-demand



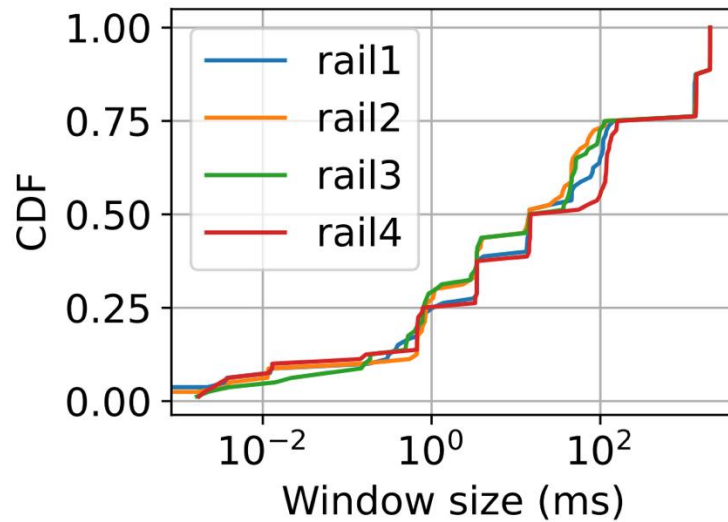
Challenge: slow reconfiguration time

~ms reconfiguration of commodity OCSes [6]

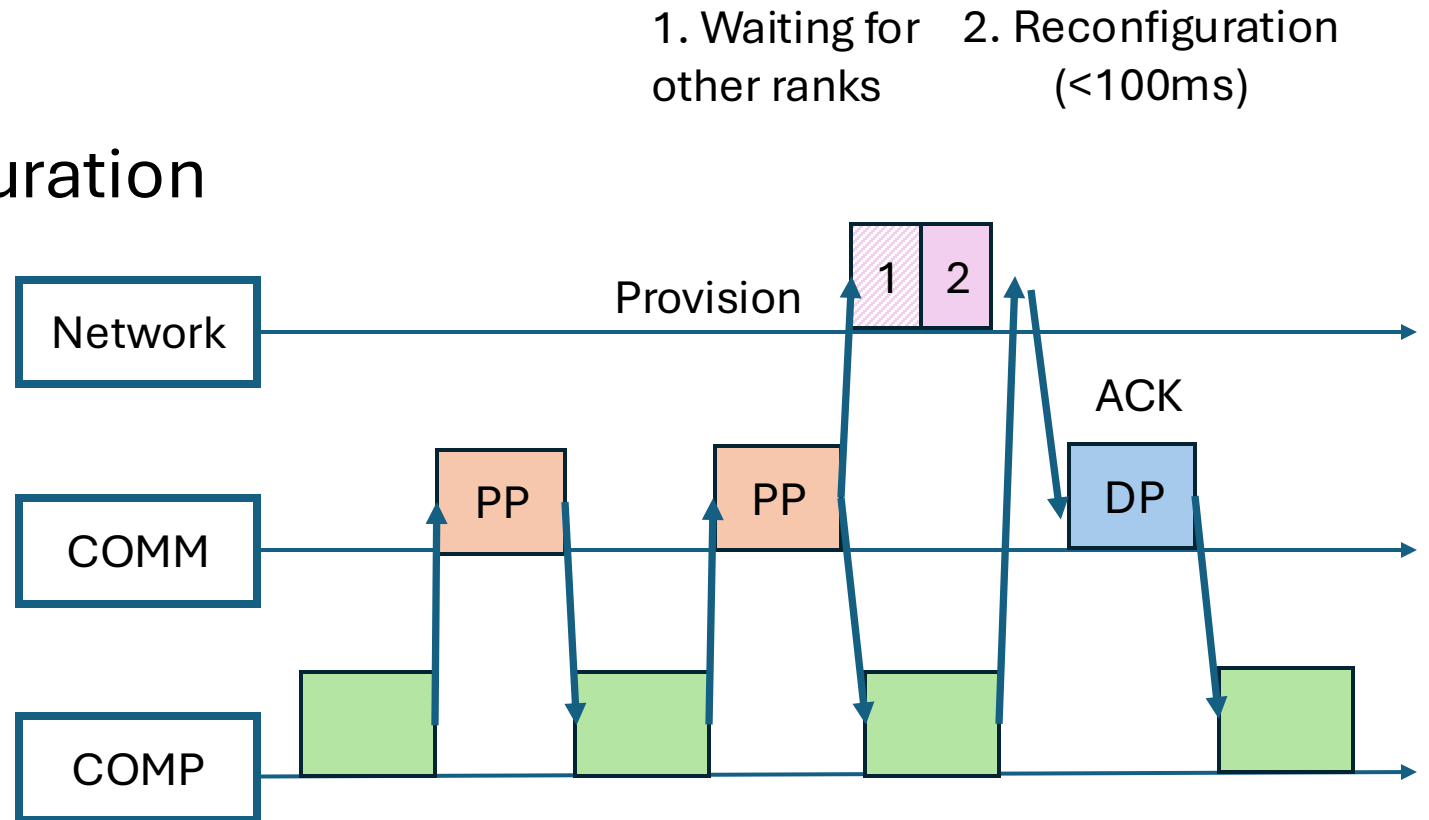


Solution: connectivity provisioning

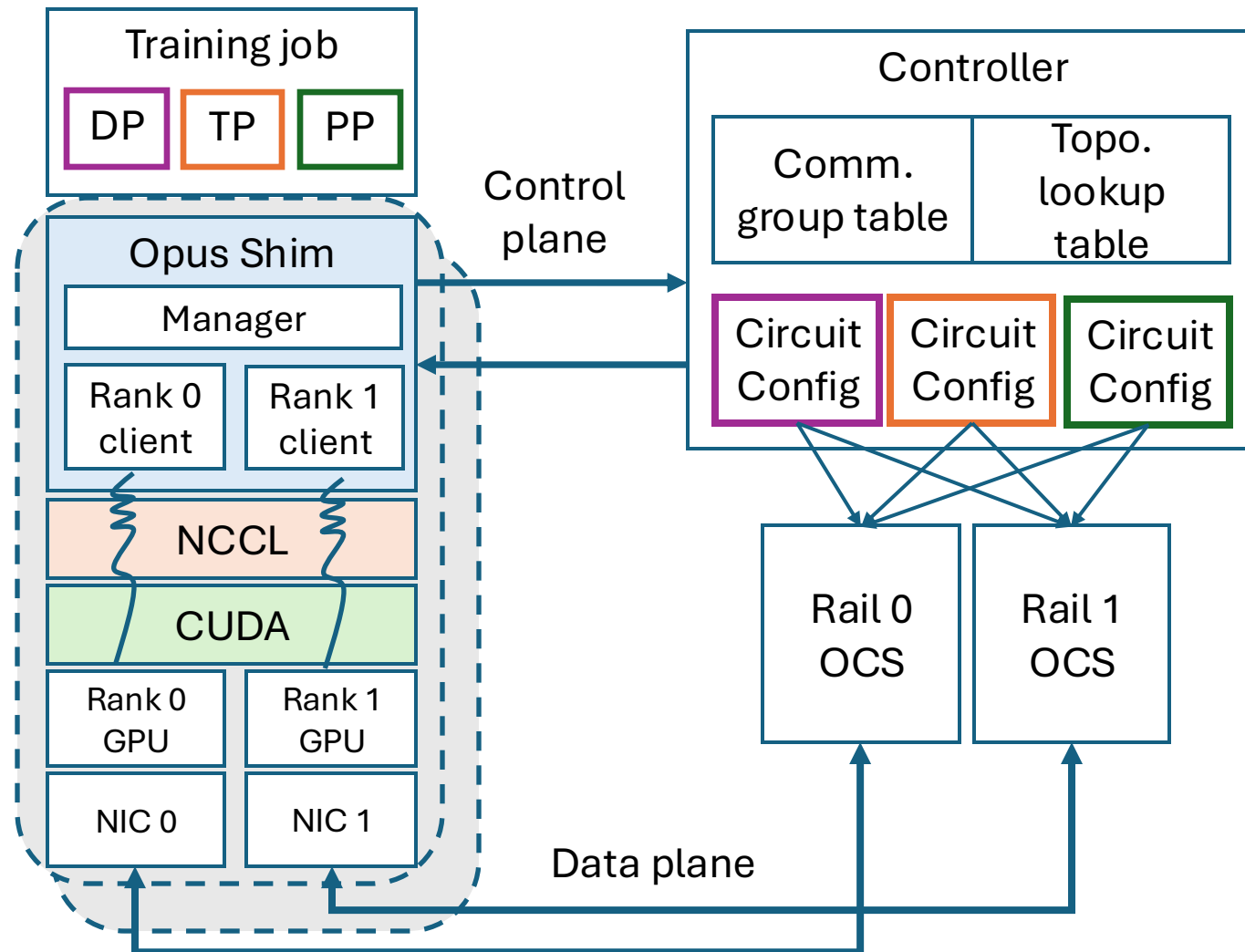
- Traffic window
- Eager topology reconfiguration



Llama-3-8B, TP=4, DP=PP=2

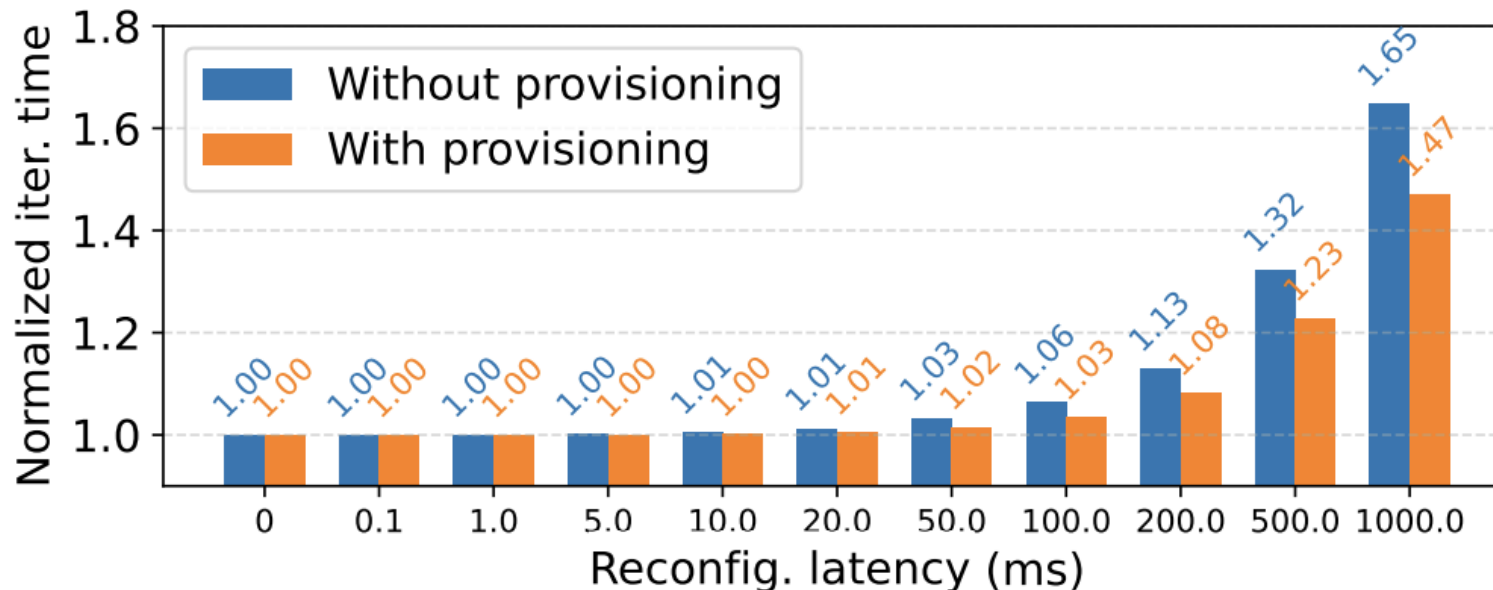


Control plane overview: Opus



Preliminary evaluation

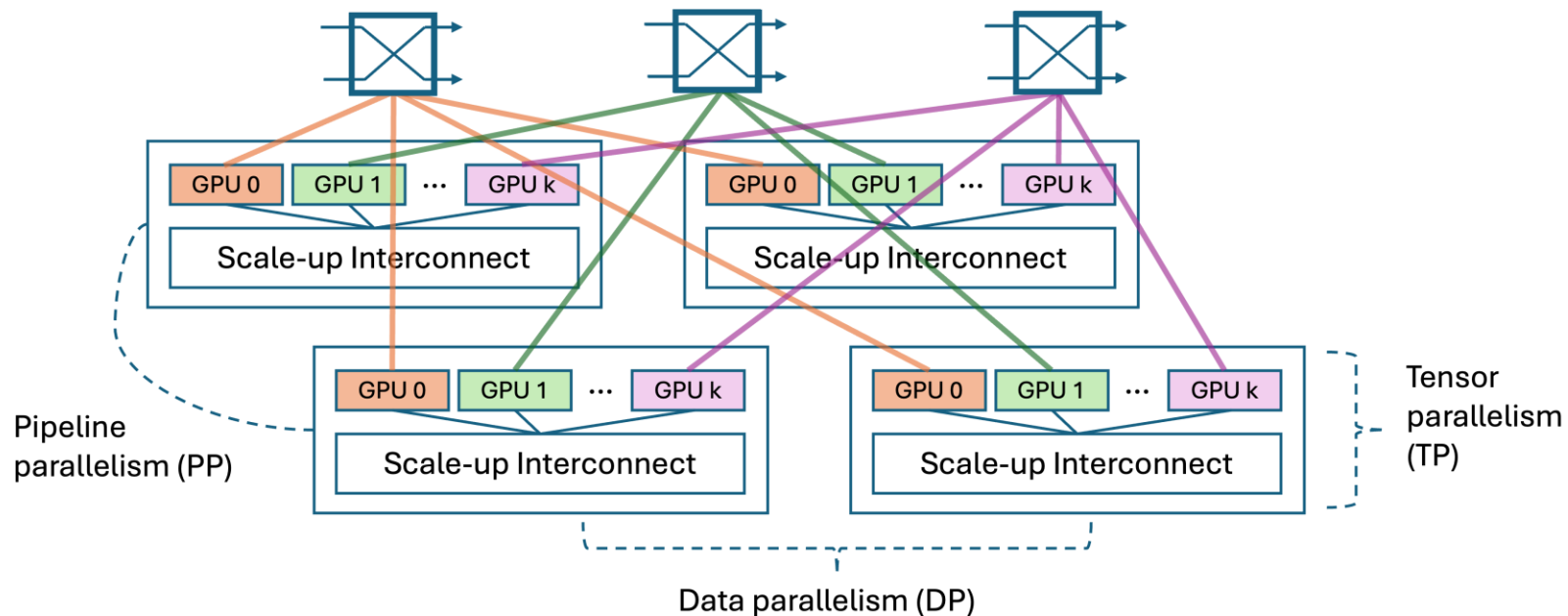
- Llama-8B model DP=PP=2, TP=4
- Emulated Iteration time for different network reconfiguration delays



Conclusion

Contact: Eric Ding, ericding@cs.cornell.edu

- Opus: control plane for in-job network reconfiguration
- Support hybrid parallelism
- Reduce network infrastructure cost and power with low overhead



Reference

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- [2] FS.COM. n.d.. N9510-64D 64-Port Ethernet L3 Data Center Switch (Broadcom Tomahawk-4, 64×400GbE). <https://www.fs.com/products/149853.html>. Accessed: 2025-07-02.
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- [6] Polatis (a HUBER+SUHNER company). n.d.. Series 7000 — 384×384-port Software-Defined Optical Circuit Switch. <https://www.polatis.com/series-7000-384x384-port-softwarecontrolled-optical-circuit-switch-sdn-enabled.asp>. Accessed: 2025-07-01.